D Karthik Reddy

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Summary

Final-year B.Tech (ECE) graduate (May 2026, CGPA: **8.97**) passionate about building **reliable, scalable, and maintainable enterprise applications**. Proficient in **Python, C, C++, JavaScript, and SQL** with hands-on experience across the full software development lifecycle – requirements analysis, design, implementation, unit testing, debugging, and code reviews. IEEE-published author, patent co-inventor, and SIH 2023 national winner. Strong communication and interpersonal skills with a self-starter mindset and the ability to thrive in fast-paced, collaborative environments.

Education

VNR Vignana Jyothi Institute of Engineering and Technology Hyderabad, India
*B.Tech in Electronics & Communication Engineering | CGPA: **8.97** | Minor in AI & ML | CGPA: **8.5*** *2022 – May 2026*

Projects

AI GitHub PR Reviewer (Live: ai-pr-reviewer-eta.vercel.app) 2026
Python | FastAPI | React | REST APIs | Git | Unit Testing | Agile

- Designed and built a **reliable, scalable application** from requirements through deployment – defined the solution scope, designed the architecture, implemented a Python backend and React frontend, and delivered iteratively in an agile workflow.
- Applied **unit testing, debugging, and structured code reviews** throughout development; performed root cause analysis on failures and applied 16 production fixes with clear documentation and descriptive Git commit history.
- Built a maintainable system that analyzed a real **Facebook React codebase** (31k stars) in **8.3 seconds**; demonstrated ability to deliver solutions that keep up with rapidly evolving requirements.

AI Email Agent (Live: ai-email-agent-alpha.vercel.app) 2026
Python | FastAPI | React | REST APIs | SQL | Git | Unit Testing

- Developed a **scalable, maintainable application** from scratch – engaged with requirements, designed a modular OOP-based architecture with 7 REST endpoints, and estimated and delivered the scope of work in iterative cycles.
- Implemented **unit and functional tests**, structured debugging, and code review practices; ensured reliability through input validation, error handling, and persistent state management with full audit logging.
- Delivered **end-to-end processing under 10 seconds** with human-in-the-loop validation; demonstrated ability to build dependable software that handles real-world complexity at scale.

Ensemble Vision Transformers – GI Disease Classification 2025
Python | PyTorch | C++ | SQL | Scikit-learn | Git | Unit Testing

- Designed and implemented a **multi-model software system** – built automated evaluation and testing pipelines, performed iterative debugging, and validated improvements using quantitative metrics across 8,000 images.
- Collaborated with team members and faculty across design and implementation; documented architecture and methodology clearly for peer review and academic submission.
- Achieved **97.75% classification accuracy**, outperforming baselines by 4.2%; research accepted at **IEEE ICIIP 2025** – demonstrating engineering rigor and strong analytical problem-solving.

UAV Magnetic Anomaly Detection – Smart India Hackathon 2023 2023
Python | C | C++ | Embedded Systems | Real-Time Systems | Git

- Led a cross-functional team to design and deliver a **real-time embedded software system** for the Ministry of Defence – engaged stakeholders, defined solutions, estimated scope, and delivered under tight constraints.
- Achieved **40% improvement in detection accuracy** and **25% increase in drone flight time**; awarded **1st place at Smart India Hackathon 2023**, Ministry of Defence track – 1M+ participants nationwide.

Technical Skills

Languages: Python, C, C++, JavaScript, SQL, HTML, CSS
Computer Science: Data Structures & Algorithms (**183 LeetCode**), Object-Oriented Design & Programming, Complexity Analysis, System Design (learning)
Software Engineering: SDLC, Agile Workflows, Unit Testing, Functional Testing, Debugging, Code Reviews, REST APIs
Web & Development: React, FastAPI, Node.js, PostgreSQL, OAuth 2.0, Git, GitHub
Core CS: Operating Systems, DBMS, Computer Networks, Distributed Systems fundamentals
AI / ML: PyTorch, TensorFlow, LangGraph, LangChain, Scikit-learn, Computer Vision

Achievements

IEEE Publication (ICIIP 2025): Lead author – Vision Transformers achieving **97.75% accuracy** on GI disease classification; peer-reviewed international conference

Smart India Hackathon 2023 – National Winner: Ministry of Defence track; 40% detection accuracy improvement, 25% flight time increase; 1M+ participants nationwide

Patent Filed: Co-inventor of AI-based acoustic real-time rail defect detection system for predictive maintenance